

In [9]:

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#Roll no :70

import      as np
import matplotlib.pyplot as plt
from sklearn.linear_model import Perceptron

# Training data
X = np.array([[1, 1], [2, 2], [3, 3], [1, 3], [2, 4], [3, 5]])
y = np.array([0, 0, 0, 1, 1, 1])

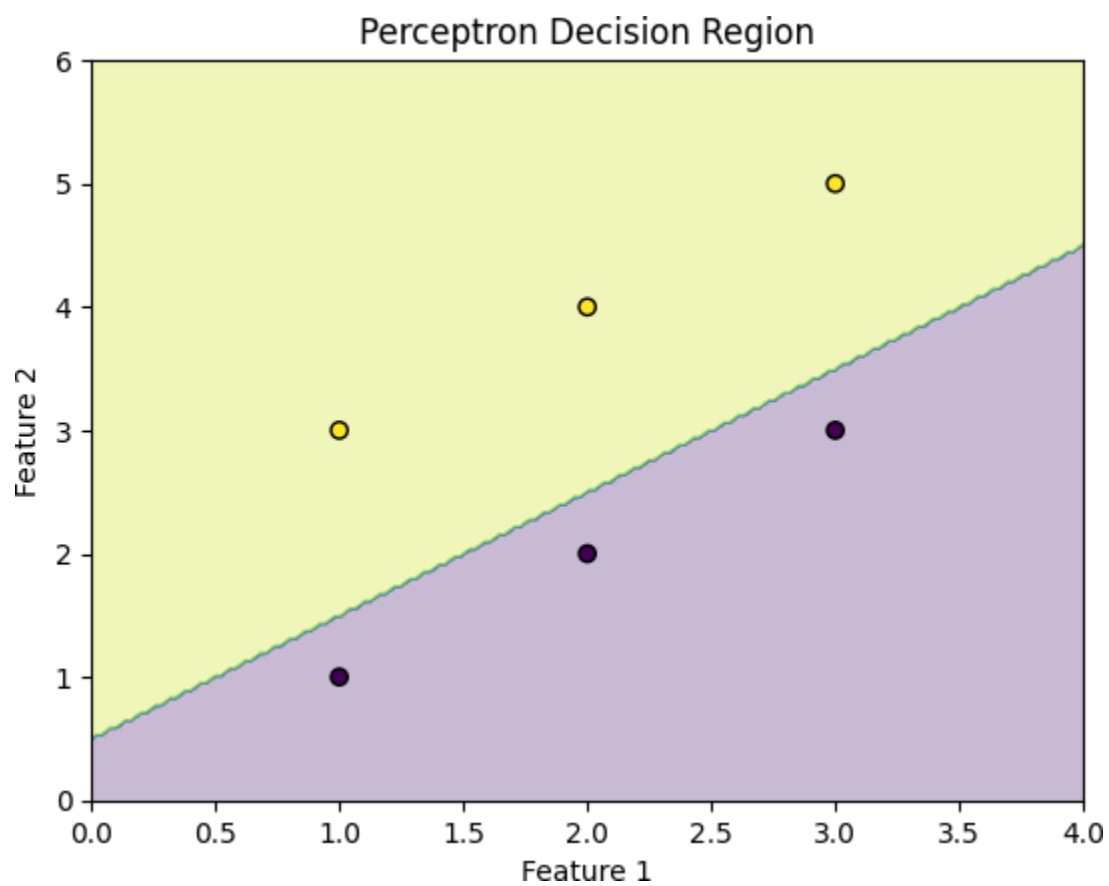
# Train perceptron
model = Perceptron(max_iter=1000, eta0=0.1)
model.fit(X, y) # ☒ Important

# Create decision region
xx, yy = np.meshgrid(
    np.linspace(0, 4, 200),
    np.linspace(0, 6, 200)
)

Z = model.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)

# Plot
plt.contourf(xx, yy, Z, alpha=0.3)
plt.scatter(X[:, 0], X[:, 1], c=y, edgecolor='k')
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("Perceptron Decision Region")

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